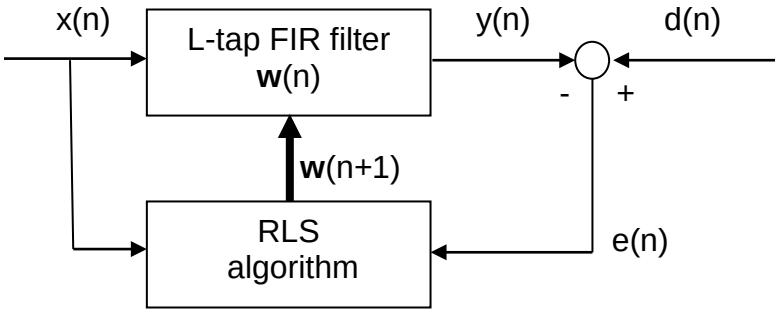
Standard RLS algorithm



Parameters

L = filter length

α = forgetting factor, $0 \ll \alpha < 1$

for "MSE" $\xi(n) = \sum_{k=0}^{n} \alpha^{n-k} e^2(k)$

Init

$$\mathbf{w}(0) = \begin{bmatrix} w_0(0) \\ \vdots \\ w_{L-1}(0) \end{bmatrix} = \mathbf{0}\,, \qquad \mathbf{x}(0) = \begin{bmatrix} x(0) \\ \vdots \\ x(-(L-1)) \end{bmatrix} = \mathbf{0}$$

$\mathbf{R}_{xx}(0) = \delta\, \mathbf{I}_L$ where $0 < \delta \ll 1 \quad \Rightarrow$

$\mathbf{R}_{xx}^{-1}(0) = \frac{1}{\delta}\, \mathbf{I}_L$

$n = 1$

Loop

Get data

$$\mathbf{x}(n) = \begin{bmatrix} x(n) \\ \vdots \\ x(n-(L-1)) \end{bmatrix}$$

$d(n)$

Predict

$$\hat{d}(n) = y(n) = \mathbf{w}^T(n-1)\, \mathbf{x}(n)$$

Error

$$e(n) = d(n) - \hat{d}(n)$$

Recursion

$$\mathbf{R}_{xx}^{-1}(n) = \frac{1}{\alpha}\left[\mathbf{R}_{xx}^{-1}(n-1) - \frac{\mathbf{R}_{xx}^{-1}(n-1)\, \mathbf{x}(n)\, \mathbf{x}^T(n)\, \mathbf{R}_{xx}^{-1}(n-1)}{\alpha + \mathbf{x}^T(n)\, \mathbf{R}_{xx}^{-1}(n-1)\, \mathbf{x}(n)}\right]$$

Update filter

$$\mathbf{w}(n) = \mathbf{w}(n-1) + \mathbf{R}_{xx}^{-1}(n)\, \mathbf{x}(n)\, e(n)$$

Iterate

$$n = n + 1$$

End

Computational complexity $\mathcal{O}(L^2)$

Performance of standard RLS algorithm

Convergence

Under stationary conditions the standard RLS algorithm will converge towards the minimum of the cost function $\xi(n) = E[e^2(n)]$ in an asymptotically unbiased way

$$E\left[\lim_{n\to\infty} \mathbf{w}(n)\right] = \mathbf{w}_{opt}$$

Convergence speed

Unlike LMS, the RLS algorithm has only one time constant of convergence towards the optimal filter coefficients

$$\tau = -\frac{1}{\log\ \alpha} = -\frac{1}{\log\ [1-(1-\alpha)]} \cong \frac{1}{1-\alpha} \qquad \text{for } 0 \ll \alpha < 1$$

This time constant can be seen to equal the effective width of the exponential window used in the definition of MSE that includes the forgetting factor, α.

Thus the convergence is – unlike LMS - not dependent on the eigenvalue spread of $\mathbf{R}_{xx}$.

In general, RLS converges must faster than LMS at the price of increased computational complexity.

Stationary and excess mean square error

As for LMS, the mean square error, $\xi = E[e^2(n)]$, will after convergence consist of two parts; one part, ξ_{min}, from the minimum MSE due to the Wiener Hopf solution, $\mathbf{w}_{opt}$, and one part, ξ_{excess}, coming from filter coefficient fluctuations away from $\mathbf{w}_{opt}$
Thus

$$\xi_{stationary} = E[e^2(n)] = \xi_{min} + \xi_{excess}$$

For RLS can be found

$$\xi_{excess} = \frac{1-\alpha}{1+\alpha} L\, \xi_{min} \ll \xi_{min} \qquad \text{for} \quad 0 \ll \alpha < 1 \text{ and moderate } L$$

The ratio between the two parts of $\xi_{stationary}$ is called the "misadjustment", $\mathcal{M}$

$$\mathcal{M} \overset{\text{def}}{=} \frac{\xi_{excess}}{\xi_{min}} \cong \frac{1-\alpha}{1+\alpha} L \ll 1 \qquad \text{for} \quad 0 \ll \alpha < 1 \text{ and moderate } L$$